

LAHORE PULSE

MASTER PRODUCT REQUIREMENTS DOCUMENT

Core Features vs Extra Features — without removing the previous PRD's detail

Smart City Hackathon Lahore — Theme 2: City Intelligence

Primary alignment: Problem Statement 1 — A Real-Time Nervous System for Lahore's Civic Data

Scope: Lahore only | **Development approach:** Vibe coding + n8n + backend + data analysis

Revision instruction: This document preserves the detailed product, data, engineering, research and trust requirements from the master PRD. The change in this revision is the classification of features into **CORE** and **EXTRA**. Details are not removed merely because a feature is extra.

1. Product Vision

Lahore Pulse is a Lahore-only city intelligence platform designed as a digital “nervous system” for the city. It connects fragmented public/open information into one live, historical and analytical view so citizens, researchers and decision-makers can understand what is happening, where it is happening, what conditions surround it, and what deserves attention.

The central product is not a generic chatbot and not a collection of unrelated cards. The system is a shared data layer: **source** → **n8n ingestion** → **validation/normalization** → **database** → **API** → **map/dashboard** → **alerts/analysis**.

The hackathon materials specifically describe Problem Statement 1 as a unified dashboard that combines two or more live data sources into a map for a district/union-council-scale civic view. The challenge also encourages multiple datasets, actionable information, maps/dashboards/apps, and architecture that can accept future official government feeds.

2. Classification Rule

CORE: features that directly establish the requested “real-time nervous system” under Problem Statement 1. These must be protected if time is limited.

EXTRA: features already defined in the product vision that make Lahore Pulse broader, more useful or more innovative, but are not necessary to prove the core problem statement.

STRETCH: advanced extras such as prediction, satellite/digital twin and advanced AI. They remain in the PRD but should only be attempted after the core is stable.

Important: “Extra” does not mean “bad” or “delete it.” It only means the team should not sacrifice the core system to build it.

3. CORE FEATURES — COMPULSORY

CORE — COMPULSORY FOR THE PROBLEM-STATEMENT-ALIGNED MVP

3.1 Unified Lahore Command Center / Home Dashboard

Why it exists: The central command view is the clearest implementation of the “nervous system” requested by Problem Statement 1.

What it does: One Lahore home screen showing current air quality, weather/rainfall, major alerts, important incidents, infrastructure context, daily changes and system freshness. It should immediately answer: What is happening now? Where? What matters?

Data / sources: OpenAQ/AQICN; Open-Meteo; OpenStreetMap; WorldPop/PBS/Open Data Pakistan; incident/report records.

How to apply: n8n feeds are normalized into one backend schema. The frontend consumes one API rather than calling every source independently. The dashboard uses reusable cards and map components.

User output: Live summary cards, alert strip, map preview, major incident feed, freshness indicators and quick links to modules.

Implementation / trust notes: Do not make this a static dashboard. Every live component should expose timestamp/source information.

3.2 Interactive Lahore Intelligence Map

Why it exists: Problem Statement 1 explicitly asks for a single map combining multiple data layers.

What it does: Lahore map with selectable layers for air stations, weather context, roads, hospitals, schools, facilities, population/context, incidents, reports and future layers.

Data / sources: OpenStreetMap; OpenAQ/AQICN; Open-Meteo; WorldPop; PBS/Open Data Pakistan; internal incident/report data.

How to apply: Use PostGIS for spatial storage/queries. OSM features can be imported through suitable extracts/Overpass/HOT/Geofabrik. Frontend renders GeoJSON/vector data with clustering for dense points.

User output: Layer controls, legend, search, zoom, clusters, selected-area summary, time filters and click-to-inspect panels.

Implementation / trust notes: Never invent an incident location. If only city-level information is known, show it as city-level rather than placing a fake pin.

3.3 Live Multi-Source Data Integration

Why it exists: The challenge requires two or more live sources and is specifically about breaking down data silos.

What it does: At minimum demonstrate two genuinely updating sources; the strongest core combination is OpenAQ + Open-Meteo, with OSM as geographic context.

Data / sources: OpenAQ/AQICN; Open-Meteo; OSM. WorldPop/PBS are slower-changing context.

How to apply: Create independent n8n workflows. Fetch → validate → normalize → store raw/normalized values → update source status → expose through API.

User output: Every feed has last-updated time, source, status and, where possible, next expected refresh.

Implementation / trust notes: Keep source adapters independent so one broken source does not break the whole dashboard.

3.4 Know My Area / Neighbourhood Intelligence

Why it exists: The challenge itself suggests a “know your neighbourhood” page. It converts the platform from a command dashboard into a citizen-facing product.

What it does: User searches/selects a Lahore area and receives a local summary: environment, weather, population context, nearby facilities, incidents and alerts.

Data / sources: OSM; OpenAQ/AQICN; Open-Meteo; WorldPop/PBS; incident/report records.

How to apply: Use PostGIS spatial queries and administrative boundaries where available. Calculate nearest facilities and counts. Associate environmental stations using defensible geographic rules.

User output: Area profile, nearby hospitals/schools, current conditions, active alerts, incident history and basic trends.

Implementation / trust notes: Clearly distinguish measured local data from city-wide or nearest-station estimates.

3.5 Roads, Hospitals, Schools, Parks & Civic Facilities

Why it exists: The challenge specifically identifies roads, hospitals, schools and facilities as valuable open map data.

What it does: Infrastructure layer that allows citizens/researchers to see what services exist around an area and relate incidents/environmental burden to infrastructure.

Data / sources: OpenStreetMap; WorldPop/PBS for population context.

How to apply: Import/query relevant OSM tags; clean duplicates; store geometry and category; use spatial joins to calculate facility access.

User output: Toggleable layers, nearest facility, facility counts and area comparisons.

Implementation / trust notes: OSM coverage is community-generated, so show the source and avoid implying official completeness.

CORE — COMPULSORY FOR THE PROBLEM-STATEMENT-ALIGNED MVP

3.6 Air Quality Intelligence

Why it exists: Air pollution is explicitly named in the challenge and gives Lahore Pulse a live environmental intelligence stream.

What it does: Current AQ observations, pollutant details, station map, AQ trend and status categories.

Data / sources: OpenAQ and/or AQICN.

How to apply: n8n scheduled polling → validate station/time/pollutant/unit → store observation → aggregate latest readings → API → frontend. Preserve historical observations.

User output: Current AQ card, station markers, pollutant details, historical chart, severity badge and stale-data warning.

Implementation / trust notes: Do not merge incompatible AQI methodologies without documenting the source/method.

CORE — COMPULSORY FOR THE PROBLEM-STATEMENT-ALIGNED MVP

3.7 Weather & Rainfall Intelligence

Why it exists: Open-Meteo is explicitly provided by the challenge and supports current, historical and forecast weather/rainfall.

What it does: Temperature, rainfall, wind and relevant forecast context for Lahore.

Data / sources: Open-Meteo.

How to apply: n8n fetches current/forecast and historical values, normalizes units and stores timestamped records.

User output: Weather card, rainfall chart, forecast, rainfall accumulation and alerts when thresholds are crossed.

Implementation / trust notes: Use weather as context for incidents and future predictive modules.

CORE — COMPULSORY FOR THE PROBLEM-STATEMENT-ALIGNED MVP

3.8 Actionable Alert Center

Why it exists: The theme asks for actionable information rather than merely displaying raw numbers.

What it does: Rule-based alerts for poor air, heavy rain/rainfall conditions, extreme heat and verified high-priority civic incidents.

Data / sources: OpenAQ/AQICN; Open-Meteo; incident/report records.

How to apply: n8n evaluates transparent rules. Store alert ID, trigger, severity, location, source, creation time, expiry and status.

User output: Critical/Warning/Watch/Info feed, map badges and area-specific alerts.

Implementation / trust notes: Rules must be explainable. Never present a model prediction as an official warning.

CORE — COMPULSORY FOR THE PROBLEM-STATEMENT-ALIGNED MVP

3.9 Source, Timestamp & Freshness System

Why it exists: Trust is essential for a live city platform. Users must know where information came from and how recent it is.

What it does: Every changing record carries source, observation time, ingestion time and freshness state.

Data / sources: All feeds.

How to apply: Database metadata fields + n8n workflow logs. Mark feeds stale after documented thresholds and expose failures.

User output: “Updated 3 min ago”, source label, stale/delayed state and source details.

Implementation / trust notes: Do not hide unavailable data. A missing feed should visibly fail gracefully.

CORE — COMPULSORY FOR THE PROBLEM-STATEMENT-ALIGNED MVP

3.10 Historical Timeline, Trends & Hotspots

Why it exists: Historical information lets the platform identify recurring problems rather than only displaying current conditions.

What it does: Time-series trends and historical event maps for environmental readings and civic incidents.

Data / sources: Stored OpenAQ/Open-Meteo data; incident/report records.

How to apply: Never overwrite observations. Python/Pandas creates daily/weekly aggregates. GeoPandas/PostGIS supports spatial hotspot analysis.

User output: Trend charts, date filters, hotspot maps and comparison between periods.

Implementation / trust notes: Historical correlations are not automatically causal relationships.

CORE — COMPULSORY FOR THE PROBLEM-STATEMENT-ALIGNED MVP

3.11 Civic Incident / Problem Intelligence

Why it exists: A nervous system should represent important real-world events when location and source evidence are available.

What it does: Accidents, road incidents, flooding/waterlogging and other civic problems with category, time, source and confidence.

Data / sources: Punjab/open datasets where available; cpo.nhmp.gov.pk / NHMP public information for relevant motorway incidents; permitted public sources; citizen reports; OSM road context.

How to apply: n8n normalizes incoming records. Store location as exact/approximate/area-only according to evidence. PostGIS links incidents to roads/areas.

User output: Incident pins/clusters, event detail, time, source, confidence and nearby context.

Implementation / trust notes: Data sources must be attributed. Do not claim a source provides live structured data unless it actually does.

CORE — COMPULSORY FOR THE PROBLEM-STATEMENT-ALIGNED MVP

3.12 Accident / Road Safety Research Foundation

Why it exists: The core system should be able to preserve incident history so the platform can later study recurring danger zones. The deeper prevention lab remains an Extra.

What it does: Basic accident history, road context and exposure information suitable for identifying candidate hotspots.

Data / sources: Accident datasets/public records; NHMP where relevant; OSM; WorldPop/PBS; weather.

How to apply: Spatially join incidents to roads, intersections, nearby schools/hospitals and population. Compute counts by road/area/time.

User output: Hotspot candidates and basic contributing-context panel.

Implementation / trust notes: Keep “correlation/context” separate from proven causes.

CORE — COMPULSORY FOR THE PROBLEM-STATEMENT-ALIGNED MVP

3.13 Citizen-Facing Search & Information Flow

Why it exists: The challenge explicitly asks for information accessible to citizens.

What it does: Simple navigation from city overview → map → selected area → event/alert → source/details.

Data / sources: All core modules.

How to apply: Frontend uses reusable components and one API. Avoid forcing users to understand the underlying datasets.

User output: Search, map, area pages, alerts, incident details and printable/shareable summaries.

Implementation / trust notes: Language/accessibility improvements can be added later without changing the backend.

CORE — COMPULSORY FOR THE PROBLEM-STATEMENT-ALIGNED MVP

3.14 Future Official-Feed Integration Architecture

Why it exists: The challenge explicitly says official systems may be closed now but the prototype should be designed so an official feed can slot in later.

What it does: Common schema and adapter architecture for future EPA, Suthra Punjab, PITB/One Map, traffic, utility or other government sources.

Data / sources: Current public sources are mapped to the same internal structures; future feeds are integration points, not falsely claimed live connections.

How to apply: n8n adapter per source → common normalized schema → same backend API → same frontend layers.

User output: Architecture diagram and optionally a clearly labelled mock feed in the demo.

Implementation / trust notes: Never claim government access that the team does not have.

4. EXTRA FEATURES — OPTIONAL, FULL DETAIL PRESERVED

These modules remain part of the product vision. They are separated only because they are not compulsory to prove Problem Statement 1. Build them after the core system is reliable.

EXTRA — OPTIONAL AFTER CORE IS COMPLETE

4.1 Daily Lahore News Intelligence

Why it exists: Citizens need daily context around events, policy, closures, development and emergencies. News is valuable but is not itself the core open-data requirement.

What it does: Daily Lahore news feed with categories, source, publication time, summary and location only where location is defensible.

Data / sources: Publisher RSS feeds/APIs and other permitted public sources; do not assume unrestricted scraping or reuse rights.

How to apply: n8n fetches feeds → deduplicates by URL/title/similarity → categorizes → optionally uses AI to summarize/classify → stores source/time.

User output: Daily Brief, breaking-news section, category filters and optional map association.

Implementation / trust notes: News should never be presented as verified government incident data unless it comes from an official source.

EXTRA — OPTIONAL AFTER CORE IS COMPLETE

4.2 Accident Hotspot & Prevention Research Lab

Why it exists: This turns historical incidents into a research tool asking: where are tragedies recurring, what conditions surround them, and what interventions should be investigated?

What it does: Hotspot maps, time-of-day/day-of-week patterns, road/intersection context, nearby schools/hospitals, weather context, population exposure and candidate intervention questions.

Data / sources: Accident/incident records; cpo.nhmp.gov.pk / NHMP public information where relevant; OSM; WorldPop/PBS; Open-Meteo.

How to apply: Python/Pandas/GeoPandas/PostGIS for spatial joins and pattern analysis. Generate transparent indicators such as incidents per road segment or exposed population, subject to data quality.

User output: Research dashboard: hotspot → context → historical trend → possible investigation questions.

Implementation / trust notes: Do not claim “X caused accidents” without a study. Phrase outputs as associations, hypotheses or research priorities.

EXTRA — OPTIONAL AFTER CORE IS COMPLETE

4.3 Citizen Reporting & Verification

Why it exists: The theme's third problem statement specifically describes turning citizens into verified sensors. Adding it gives Lahore Pulse a way to generate new civic data.

What it does: Citizen submits issue with location, timestamp, category and optional photo/evidence. Public record shows open/resolved status.

Data / sources: OpenStreetMap for basemap/boundaries; platform-generated data.

How to apply: Frontend report form → backend record → n8n classification/duplicate candidate → moderation/verification → public status.

User output: Map pins, public issue feed, status timeline, duplicate candidates and evidence.

Implementation / trust notes: Verification must be labelled clearly; an unverified citizen report is not equivalent to an official report.

EXTRA — OPTIONAL AFTER CORE IS COMPLETE

4.4 Duplicate Detection & Evidence Layer

Why it exists: Multiple citizens may report the same incident. Deduplication makes citizen-generated data more useful.

What it does: Identify reports that are close in space/time and similar in category/text/photo metadata.

Data / sources: Citizen reports; OSM geometry.

How to apply: Use PostGIS distance/time windows plus text/embedding similarity if useful. Route candidates to moderation rather than automatically deleting.

User output: “Possible duplicate” grouping and combined report count.

Implementation / trust notes: False merges can be harmful; keep original reports accessible.

EXTRA — OPTIONAL AFTER CORE IS COMPLETE

4.5 Flood / Waterlogging Risk

Why it exists: Lahore can benefit from early indication of rainfall-related local risk, especially when combined with terrain and drainage context.

What it does: Area risk score based on recent/forecast rainfall and physical context, with citizen reports for validation.

Data / sources: Open-Meteo; elevation/DEM where legally/technically accessible; OSM drainage/roads; citizen reports; WorldPop.

How to apply: Start with an explainable rule-based score. Later train/validate a model if sufficient historical labels exist.

User output: Risk map, score components, rainfall forecast and affected population context.

Implementation / trust notes: Do not call it a flood forecast unless the model is actually validated for that task.

EXTRA — OPTIONAL AFTER CORE IS COMPLETE

4.6 Heat Exposure Intelligence

Why it exists: Temperature alone does not show who is exposed. Combining environmental conditions with population improves civic usefulness.

What it does: Heat burden/exposure by Lahore area, emphasizing population and potentially land-cover/surface-temperature context.

Data / sources: Open-Meteo; WorldPop; Copernicus/GEE where feasible.

How to apply: Calculate area-level exposure indicators; optionally integrate satellite-derived surface temperature.

User output: Heat map, population-exposure score and vulnerable-area ranking.

Implementation / trust notes: Surface temperature and air temperature are different measurements and must not be presented as interchangeable.

EXTRA — OPTIONAL AFTER CORE IS COMPLETE

4.7 Petrol Price Intelligence

Why it exists: Petrol prices affect daily Lahore life and can make the platform useful beyond emergencies, but they are not a direct Problem Statement 1 requirement.

What it does: Current effective petrol/diesel prices, previous price, change and effective date.

Data / sources: Official Pakistan petroleum price announcements and other authoritative public sources; verify current availability before implementation.

How to apply: n8n checks source → extracts effective price/date → stores history → calculates change → publishes card.

User output: Petrol card, historical trend and price-change alert.

Implementation / trust notes: Use effective dates and source attribution; do not invent station-level prices.

EXTRA — OPTIONAL AFTER CORE IS COMPLETE

4.8 Essential Food & Vegetable / DC Rates

Why it exists: Daily food prices are highly relevant to citizens and can be presented as a city cost-of-living layer.

What it does: Fruits, vegetables, poultry and grains with unit and daily/official rate.

Data / sources: Qeemat Punjab/PITB or authoritative Lahore/Punjab public rate sources.

How to apply: n8n retrieves structured data where available or processes permitted public rate publications; normalize product/unit/date.

User output: Daily price board, changes and history.

Implementation / trust notes: Keep official/DC rate separate from observed market/shop prices.

EXTRA — OPTIONAL AFTER CORE IS COMPLETE

4.9 Gold & Silver Rates

Why it exists: Useful daily financial information for Lahore residents, but unrelated to the core civic command requirement.

What it does: 24K/22K gold and silver rates per tola/gram with date and change.

Data / sources: APGJA/public rate sheets or a permitted API such as a suitable Pakistan gold-rate provider; verify licensing/availability.

How to apply: n8n fetch/parse → normalize units/purity → store history → compare with previous rate.

User output: Gold/silver cards and history.

Implementation / trust notes: Source and publication time should be displayed.

EXTRA — OPTIONAL AFTER CORE IS COMPLETE

4.10 Currency Exchange Rates

Why it exists: Useful economic information and a practical daily module.

What it does: USD, SAR, AED and other selected currencies against PKR; distinguish interbank/open-market when data supports it.

Data / sources: SBP public data and permitted FX APIs; ECAP/public sources where accessible.

How to apply: n8n fetch → normalize market/category → timestamp → store → calculate changes.

User output: FX cards, change and history.

Implementation / trust notes: Never present a global mid-market rate as Pakistan open-market rate.

EXTRA — OPTIONAL AFTER CORE IS COMPLETE

4.11 PSX / Market Snapshot

Why it exists: Adds financial/economic intelligence but is not necessary for city intelligence.

What it does: KSE-100/KMI-30, selected equities, gainers/losers and volume during market hours.

Data / sources: PSX public/licensed data or a permitted market provider. Candidate APIs such as Capital Stake/Twelve Data require current coverage/terms verification.

How to apply: n8n/API integration → normalize ticker/index/session/time → store snapshots.

User output: Market cards, top movers and index trend.

Implementation / trust notes: Do not assume a free API provides real-time PSX data until verified.

EXTRA — OPTIONAL AFTER CORE IS COMPLETE

4.12 LPG & Gas Cylinder Prices

Why it exists: Useful household energy-cost information.

What it does: Official LPG price per kg/cylinder and effective period.

Data / sources: OGRA public notifications.

How to apply: n8n monitors permitted official publications → extracts effective rates → stores history.

User output: LPG price card and change alert.

Implementation / trust notes: Keep official notified price separate from retail seller price.

EXTRA — OPTIONAL AFTER CORE IS COMPLETE

4.13 Power Outage / LESCO Alerts

Why it exists: Power disruption is highly relevant to Lahore citizens, but public machine-readable coverage may be limited.

What it does: Scheduled shutdowns, maintenance notices and accessible outage information.

Data / sources: LESCO public notices/portal or another authorized public feed. K-Electric/FESCO are examples from the broader source list but are not Lahore sources and should not be used as Lahore evidence.

How to apply: n8n monitors accessible source → parses feeder/area/time → creates alert record.

User output: Power alerts by area and scheduled maintenance.

Implementation / trust notes: Do not claim live feeder status if the source is not actually live/structured.

EXTRA — OPTIONAL AFTER CORE IS COMPLETE

4.14 Traffic Diversions, Closures & Advisories

Why it exists: Road closures and diversions are directly useful to citizens and can be connected to OSM.

What it does: Rallies, closures, diversions, VIP-route advisories, fog/highway warnings and other officially published notices.

Data / sources: City Traffic Police Lahore/public advisories; NHMP advisories for highways/motorways; OSM.

How to apply: n8n monitors public advisories → extracts named roads/time → manually/algorithmically matches road geometry → publishes temporary map layer.

User output: Temporary closure/diversion overlays and alerts.

Implementation / trust notes: Do not claim live congestion without a real traffic feed. The challenge explicitly warns that live Lahore traffic feeds are not openly available.

EXTRA — OPTIONAL AFTER CORE IS COMPLETE

4.15 Flight & Train Status

Why it exists: Useful travel information for Lahore residents but outside the core civic intelligence requirement.

What it does: Arrivals/departures, delays and status for Lahore Airport and Lahore Railway where data is legitimately accessible.

Data / sources: CAA/airport public information, Pakistan Railways public systems, or licensed providers such as Flightradar24 where permitted.

How to apply: n8n/API ingestion → normalize flight/train identifiers, scheduled/actual time and status.

User output: Transport status panels and delay indicators.

Implementation / trust notes: Provider terms and real-time availability must be verified before implementation.

EXTRA — OPTIONAL AFTER CORE IS COMPLETE

4.16 Infrastructure Growth & Service-Gap Analysis

Why it exists: Researchers and planners can compare population, services, roads and development to identify underserved areas.

What it does: Service density, population exposure, facility access and potentially development/land-cover change.

Data / sources: OSM; WorldPop; PBS; Open Data Pakistan; Copernicus/GEE.

How to apply: Use spatial joins and area-level indicators. Compare facilities/population rather than claiming service quality from presence alone.

User output: Area ranking, service-gap map and research view.

Implementation / trust notes: Missing OSM features do not necessarily mean a real facility is absent.

EXTRA — OPTIONAL AFTER CORE IS COMPLETE

4.17 Satellite / Small Digital Twin

Why it exists: Provides a visually strong representation of one Lahore neighborhood and supports scenario analysis.

What it does: Buildings/roads + land cover + surface heat/water/greenery + population for one defined neighborhood.

Data / sources: OSM; Copernicus; Google Earth Engine; WorldPop; Open-Meteo.

How to apply: Select a small area, process satellite layers, combine with OSM and expose scenario toggles.

User output: 3D/simplified digital twin, heat/greenery/water scenario.

Implementation / trust notes: Do not attempt a whole-city digital twin during a short hackathon.

EXTRA — OPTIONAL AFTER CORE IS COMPLETE

4.18 Predictive Air Quality

Why it exists: The theme encourages prediction. A validated local forecast can turn monitoring into advance warning.

What it does: Forecast air quality for a selected station/area using historical pollution and weather features.

Data / sources: OpenAQ/AQICN history + Open-Meteo history/forecast.

How to apply: Python feature engineering → time-based train/validation split → model → forecast API. Report error metrics and confidence/limitations.

User output: Forecast chart, predicted category and uncertainty/error information.

Implementation / trust notes: Prediction quality depends heavily on historical station coverage; do not overclaim neighborhood-level accuracy.

EXTRA — OPTIONAL AFTER CORE IS COMPLETE

4.19 AI Intelligence Layer

Why it exists: AI is useful when applied to repetitive intelligence tasks, but a generic chatbot is not necessary for a Lahore-only platform.

What it does: News classification/summaries, incident categorization, duplicate candidates, anomaly explanations and structured daily brief generation.

Data / sources: Lahore Pulse's own verified/attribution data; selected permitted text sources.

How to apply: n8n triggers AI workflows only after data validation. Store the underlying records supporting each generated insight.

User output: AI insight cards with links to underlying source records.

Implementation / trust notes: AI should not invent locations, incidents, statistics or official warnings.

5. Data Source Register

Core sources: OpenAQ / AQICN for air quality; Open-Meteo for weather/rainfall; OpenStreetMap for roads/facilities; WorldPop and PBS/Open Data Pakistan for population/demographic context; available public accident/incident sources including relevant NHMP information.

Extra sources: public/official Lahore news feeds; Qeemat Punjab/PITB; OGRA; SBP/ECAP or permitted FX providers; PSX or licensed market providers; LESCO public notices; Lahore Traffic Police; NHMP; Pakistan Railways/aviation sources; APGJA/public bullion rates.

Satellite: Copernicus Browser and Google Earth Engine where terms and hackathon use permit.

Every source must be checked for current accessibility, rate limits, attribution requirements, terms of use and whether the data is genuinely live, historical or static before being presented as such.

6. Data Acquisition & n8n Architecture

n8n is the orchestration layer. Each source gets its own workflow so failures are isolated.

- Trigger: schedule/webhook/manual refresh.
- Fetch: API, RSS, permitted webpage extraction or file source.
- Validate: required fields, timestamp, units, coordinates and source response.
- Normalize: convert to Lahore Pulse common schema.
- Deduplicate: prevent repeated records where source identifiers allow it.
- Enrich: geocode/spatially associate only when defensible.
- Store: raw payload reference + normalized record + source metadata.
- Analyze: rules, aggregation, Python/Pandas/GeoPandas jobs or AI workflows.
- Serve: backend API.
- Display: frontend map/dashboard.
- Monitor: workflow success, failures, stale data and source health.

Key rule: the frontend should not contain source-specific business logic. Adding a new government feed later should require an adapter/workflow and schema mapping, not a redesign of the UI.

7. Backend / Database Requirements

Suggested stack: PostgreSQL + PostGIS; FastAPI or equivalent backend; n8n for ingestion/orchestration; Python/Pandas/GeoPandas for analysis; modern web frontend with a map library.

Core entities: sources, source_runs, observations, air_quality_readings, weather_readings, facilities, roads, areas, incidents, alerts, citizen_reports, news_items, price_records and analysis_results.

Every dynamic record should support: id, source_id, observed_at, ingested_at, location/geometry where applicable, category/type, value/payload, confidence/status and metadata.

Spatial requirements: use PostGIS for distance, intersection, area aggregation, nearest-facility and hotspot calculations. Keep Lahore boundary and relevant administrative/OSM geometries centrally available.

8. Frontend / Backend Developer Merge

The frontend developer should build against a documented API contract rather than directly integrating every data source. The backend developer owns database, n8n and API. Both developers use Git/GitHub with agreed branch naming and API schemas.

Recommended sequence: backend defines sample JSON/API contracts first → frontend builds against mock data → backend replaces mocks with live data → integration testing → deployment.

This allows two developers to work in parallel without repeatedly merging source-specific code into the frontend.

9. Data Trust, Location & Verification Rules

- Never invent an exact location from a general news report.
- Separate official, public-source, citizen-reported and AI-derived information.
- Show source and timestamp for changing information.
- Use approximate/area-level location when exact coordinates are unavailable.
- Never convert an unverified report into a verified fact.
- Do not call a rule-based threshold a machine-learning prediction.
- Do not claim live traffic, utility or government data unless the actual source provides it.
- Preserve raw/source metadata so an analyst can trace an output back to its input.
- AI summaries must remain linked to underlying records.

10. Success Criteria

- At least two independent live open data sources are visibly working during the demo.
- A user can understand Lahore conditions from one map/dashboard without visiting multiple source websites.
- Selecting an area provides meaningful local context.
- Every live value has source/freshness information.
- At least one actionable alert is generated from a transparent rule.
- Historical records are retained and can produce a trend/hotspot view.
- Incident information is location-aware only where supported by evidence.
- The architecture can accept a future official feed without changing the core frontend.
- Extra modules never compromise the reliability of the core.

11. Recommended Build Priority

Priority	Build	Status
1	Database + Lahore spatial foundation + API contract	CORE
2	OSM map + infrastructure layers	CORE
3	OpenAQ/AQICN ingestion	CORE
4	Open-Meteo ingestion	CORE
5	Unified dashboard + Know My Area	CORE
6	Alerts + source/freshness	CORE
7	Historical trends + incident layer	CORE
8	Polish, testing, demo and future-feed architecture	CORE
9	Accident prevention research lab	EXTRA / High value
10	Daily Lahore news	EXTRA / High value
11	Petrol + food prices + traffic/utilities	EXTRA / High value
12	Citizen reporting	EXTRA
13	Gold/FX/PSX/LPG/transport	EXTRA
14	Flood/heat prediction, satellite, digital twin, advanced AI	STRETCH

12. Final Product Definition

The finished Lahore Pulse should feel like one connected city nervous system rather than a dashboard containing unrelated widgets. The core observes Lahore through multiple live/open sources, connects those sources spatially and temporally, explains what is happening, alerts citizens to meaningful changes, preserves history, and creates a foundation for research and future government integration.

The extras then extend that same nervous system into daily news, accident prevention research, prices, transport, utilities, citizen reports, prediction and advanced analysis. **Nothing from the previous detailed PRD is removed merely because it is classified as Extra.** The classification only determines build priority.